TAYLOR PENTZ

Senior Python AI/ML & LLM Engineer
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Professional Summary

Senior **Python AI/ML and LLM Engineer** with 10+ years delivering enterprise SaaS, healthcare, fintech, and analytics platforms using **Python 3.8–3.12**, **FastAPI**, **Django**, **PyTorch 1.13–2.x**, **TensorFlow 2.x**, **LangChain**, **RAG**, vector databases, and **AWS/GCP**, converting complex data and language workflows into secure, observable, cost-aware products that improve automation, decision support, and engineering velocity.

Technical Skills

- **Languages: Python 3.8–3.12**, SQL, JavaScript, TypeScript, Bash
- **AI / ML / LLM: LangChain**, **LangGraph**, **RAG**, prompt engineering, embeddings, model evaluation, fine-tuning basics, NLP pipelines, semantic search, feature engineering, recommendation systems
- **ML Frameworks: PyTorch 1.13–2.x**, **TensorFlow 2.x**, scikit-learn, XGBoost, pandas, NumPy, Hugging Face Transformers
- **Backend Engineering: FastAPI**, Django, Flask, REST APIs, GraphQL basics, async APIs, background jobs, microservices, event-driven architecture
- **Vector / Search: pgvector**, Pinecone, OpenSearch, Elasticsearch, FAISS, hybrid search, metadata filtering, reranking workflows
- **Cloud Platforms: AWS** Lambda, ECS, S3, RDS, SQS, SNS, CloudWatch, Bedrock; **GCP** Cloud Run, BigQuery, Vertex AI, Cloud Storage; Azure OpenAI
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, SQL Server, DynamoDB basics
- **MLOps / DevOps:** Docker, Kubernetes, GitHub Actions, GitLab CI, Jenkins, MLflow, model registry concepts, CI/CD, Terraform basics
- **Testing & Quality:** pytest, unittest, integration testing, API contract testing, model regression tests, golden datasets, load testing
- **Monitoring:** OpenTelemetry, Prometheus, Grafana, CloudWatch, structured logging, distributed tracing, alerting, incident review
- **Engineering Practices:** Clean Architecture, domain-driven design, secure API design, code review, technical mentoring, production debugging, Agile/Scrum

Professional Experience

SumatoSoft

Senior Software Engineer
Boston, MA (Remote) | Apr 2023 – Mar 2025

- Built production **LLM and NLP services** with **Python 3.10–3.11**, **PyTorch 1.13–2.x**, **Hugging Face Transformers**, **LangChain**, and **FastAPI**, supporting summarization, document classification, semantic search, and retrieval-augmented generation workflows.
- Developed model evaluation pipelines with **pandas**, **NumPy**, **scikit-learn**, and custom benchmark datasets, helping compare embedding quality, retrieval accuracy, prompt changes, and model regression before production releases.

- Implemented classification and ranking experiments using **XGBoost**, **scikit-learn**, and transformer embeddings, improving document routing accuracy and reducing manual review effort by **31%** for business users.
- Optimized inference workflows for **PyTorch** and transformer-based models by applying batching, caching, smaller model routing, and async API execution, reducing average AI response latency by **38%** during peak usage.
- Designed production **LLM orchestration** patterns with prompt templates, retrieval guards, retry policies, token budgeting, and response validation, reducing hallucination-related support tickets by **31%** across client-facing assistants.
- Built embeddings pipelines using **OpenAI**, **Azure OpenAI**, **Sentence Transformers**, **pgvector**, and **Pinecone**, improving semantic search relevance for policy, contract, and support content while keeping ingestion jobs replayable.
- Modernized legacy Python APIs from synchronous request handlers into async **FastAPI** services with **Pydantic**, background workers, and structured error responses that improved endpoint latency by **38%** under peak document-analysis traffic.
- Resolved production issues involving malformed PDF extraction, chunk overlap drift, duplicate embeddings, timeout-prone completions, and stale cache reads by adding validation checkpoints, idempotency keys, and observability spans.
- Implemented model evaluation workflows with golden datasets, regression checks, cosine-similarity thresholds, and human review queues so product teams could compare prompt and retrieval changes before deployment.
- Created **MLOps** deployment flows with **Docker**, **GitHub Actions**, **AWS ECS/Lambda**, **GCP Cloud Run**, and environment-based configuration, allowing safer releases across staging and production AI services.
- Optimized inference and retrieval costs through batch embedding, cache-first retrieval, prompt compression, and model routing between larger and smaller LLMs, lowering monthly AI infrastructure spend by **24%**.
- Partnered with product, data, and security stakeholders to add role-aware document permissions, audit logging, PII masking, and tenant isolation for enterprise customers using regulated knowledge bases.
- Mentored engineers on **Python architecture**, **LLM testing**, code review quality, and incident analysis while documenting reusable patterns for retrieval services, background jobs, and API contracts.

Hexaview Technologies Inc.

Software Engineer
New York, NY | Jul 2018 – Mar 2023

- Built machine learning pipelines with **Python 3.7–3.9**, **pandas**, **NumPy**, **scikit-learn**, and **XGBoost** for classification, scoring, and recommendation features across analytics-heavy business applications.
- Developed early deep learning and NLP workflows with **TensorFlow 2.x** and **Hugging Face Transformers**, supporting text similarity, entity extraction, intent classification, and automated document tagging features.
- Improved feature-engineering quality by standardizing **pandas** transformation logic, missing-value handling, outlier detection, label encoding, and train-test validation across multiple ML services.
- Created model-serving APIs using **Flask**, **Django**, **Celery**, and **PostgreSQL**, allowing frontend and reporting systems to consume prediction scores, confidence values, and explainability metadata.
- Introduced REST APIs for model inference, batch scoring, and experiment metadata, giving frontend and reporting teams consistent access to predictions, confidence scores, and failure states.
- Migrated older Python services from loosely structured scripts into layered application modules with service classes, repository patterns, typed DTOs, and reusable validation logic.
- Resolved recurring production defects around Celery retry storms, missing timezone normalization, CSV parsing failures, and slow ORM queries by adding dead-letter handling, indexes, and stronger input contracts.

- Improved data pipeline reliability by building incremental extraction jobs, checksum validation, backfill commands, and alerting around failed loads in mixed **AWS** and on-premise environments.
- Implemented CI pipelines with **Jenkins**, **GitLab CI**, **Docker**, unit tests, integration tests, and deployment gates that shortened release verification time by **34%** for Python services.
- Supported early NLP use cases with tokenization, keyword extraction, text similarity, and entity cleanup, creating foundations that later made transformer-based search and LLM integrations easier to adopt.
- Collaborated with solution architects and business analysts to translate ambiguous reporting and automation needs into maintainable APIs, database schemas, model features, and production runbooks.

SimbirSoft Company

Software Developer
Boston, MA | Mar 2015 – Jun 2018

- Supported data-heavy backend workflows with **Python**, **NumPy**, **pandas**, **Django**, and **PostgreSQL**, building reporting, import validation, and operational analytics features for client-facing systems.
- Used **scikit-learn** for practical classification and scoring prototypes, including duplicate detection, data quality checks, and rule-assisted recommendations for internal workflow automation.
- Improved legacy reporting performance by replacing repeated raw SQL and manual spreadsheet logic with reusable **pandas** transformations, indexed queries, and cleaner service-layer processing.
- Delivered automation modules for data imports, validation workflows, notification rules, and reporting exports that reduced repetitive back-office processing by **22%** for client operations teams.
- Refactored legacy monolithic views, raw SQL blocks, and duplicated business rules into reusable service functions, cleaner serializers, and safer transaction boundaries.
- Fixed production bugs involving Unicode encoding errors, broken file uploads, session expiration loops, incorrect pagination totals, and race conditions in status updates.
- Supported gradual modernization from older Python and Django patterns by separating configuration, improving dependency pinning, and preparing services for Python 3 compatibility.
- Created practical test coverage with **unittest**, **pytest**, mocked service calls, and regression cases for critical workflows such as order updates, billing exports, and customer notifications.
- Worked closely with QA, project managers, and frontend developers to ship new workflow features, investigate client-reported defects, and keep delivery predictable across multiple concurrent projects.

Microsoft

Software Developer Intern
Cambridge, MA | Aug 2014 – Feb 2015

- Contributed as a **Software Developer Intern** on internal engineering tools using **Python 2.7**, **C#**, **SQL Server**, and early **Azure** services for data collection and operational reporting.
- Built scripts and small services that normalized telemetry records, flagged incomplete payloads, and helped senior engineers inspect reliability patterns across internal application components.
- Assisted with prototype ML-style analysis for log categorization and anomaly review, using Python data-processing libraries to identify repeated error signatures and noisy alert groups.
- Improved SQL queries, stored procedures, and data validation checks used by reporting dashboards, making internal metrics easier to reproduce during investigation sessions.

- Fixed intern-level defects related to null handling, incorrect joins, failed scheduled jobs, and inconsistent timestamp formatting across development and test environments.
- Participated in code reviews, sprint planning, debugging sessions, and documentation updates while learning enterprise engineering practices around source control and release discipline.
- Created clear handoff notes, setup instructions, and test cases for assigned tools so full-time engineers could maintain the work after the internship ended.

Projects

Enterprise RAG Knowledge Assistant

- Designed a production **RAG-based assistant** for enterprise document search using **Python**, **FastAPI**, **LangChain**, **OpenAI/Azure OpenAI**, **pgvector**, **PostgreSQL**, and **Redis**, supporting document ingestion, chunking, embeddings, semantic retrieval, source citation, role-based access, and evaluation workflows for regulated internal knowledge bases.

ML Forecasting and Decision Support Platform

- Built a **machine learning decision-support platform** using **Python**, **scikit-learn**, **TensorFlow 2.x**, **FastAPI**, **Celery**, **PostgreSQL**, **Docker**, and **AWS**, delivering batch scoring, feature validation, model monitoring, API-based predictions, and reporting outputs for operational teams managing high-volume business records.

Certifications

- Google Cloud Professional Machine Learning Engineer
- Microsoft Certified: Azure AI Engineer Associate
- AWS Certified Solutions Architect – Associate
- Databricks Certified Machine Learning Professional

Education

University of North Carolina at Chapel Hill

Bachelor’s Degree in Computer Science (2010 – 2014)